

Weekly Report 9: Complete Pipeline from Image Loading to 3D OBJ Formation

Project Number: 18

1 Introduction

This week, we successfully created a **complete pipeline** that takes diamond images as input and produces a 3D OBJ file as output.

After testing our visual hull carving with different angle images in the previous week, we integrated all the components into a single, working pipeline that covers the entire journey from image loading to 3D model formation.

2 Methodology

Our complete pipeline consists of the following stages:

1. **Image loading** — reading all diamond silhouette images from the dataset.
2. **Silhouette extraction** — converting images to binary masks.
3. **Voxel grid creation** — building a 3D voxel grid.
4. **Visual hull carving** — removing voxels outside the silhouettes.
5. **Mesh extraction** — converting voxels to a triangular mesh.
6. **3D OBJ export** — saving the final model as an OBJ file.

2.1 Complete Pipeline Implementation

We created a single function that combines all stages of the reconstruction process:

Implementation in Python:

```
def complete_pipeline(image_folder, output_file="diamond_model.obj"):
    # Step 1: Load all images
    images = load_all_images(image_folder)

    # Step 2: Extract silhouettes
    silhouettes = []
    for img in images:
        silhouette = extract_silhouette(img)
        silhouettes.append(silhouette)
```

```

# Step 3: Create voxel grid
voxel_grid = create_voxel_grid(resolution=64)

# Step 4: Visual hull carving
carved_grid = carve_visual_hull(voxel_grid, silhouettes)

# Step 5: Extract mesh
from skimage import measure
vertices, faces, normals, _ = measure.marching_cubes(
    carved_grid, level=0.5
)

# Step 6: Export to OBJ
export_to_obj(vertices, faces, output_file)

print(f" 3D model saved to {output_file}")

# Run the complete pipeline
complete_pipeline("diamond_images/", "diamond_model.obj")

```

2.2 Pipeline Stages

Each stage of the pipeline performs a specific task:

- **Image loading:** Reads all BMP images from the dataset folder, handling different file names and formats.
- **Silhouette extraction:** Converts each image to a binary mask where white pixels represent the diamond and black pixels represent the background.
- **Voxel grid creation:** Builds a 3D grid of voxels (we use 64^3 based on our previous optimization).
- **Visual hull carving:** Projects each voxel onto all silhouette images and removes voxels that fall outside any silhouette.
- **Mesh extraction:** Converts the remaining voxels into a triangular mesh using the Marching Cubes algorithm.
- **OBJ export:** Saves the mesh as a standard OBJ file that can be opened in any 3D software.

3 Results

After implementing the complete pipeline, we successfully generated a 3D OBJ file from our diamond images.

Key achievements:

- Successfully created a complete pipeline that processes images from start to finish.

Table 1: Complete Pipeline Performance

Stage	Time	Output
Image loading	2.1 seconds	200 images loaded
Silhouette extraction	3.5 seconds	200 binary masks
Voxel grid creation	0.8 seconds	64^3 grid
Visual hull carving	12.3 seconds	Carved voxel model
Mesh extraction	0.9 seconds	Triangular mesh
OBJ export	0.2 seconds	diamond_model.obj
Total	19.8 seconds	Complete 3D model

- The pipeline takes diamond images as input and produces a 3D OBJ file as output.
- Total processing time is under 20 seconds for 200 images.
- The exported OBJ file can be opened in standard 3D software (Blender, MeshLab, etc.).
- The final 3D model accurately represents the diamond’s structure.

4 Discussion

Creating the complete pipeline helped us understand the entire 3D reconstruction process:

- **Integration is key** — combining all stages into a single pipeline makes the process much easier to use and test.
- **Each stage is important** — from image loading to OBJ export, every step contributes to the final result.
- **Processing time** is reasonable — the entire pipeline completes in under 20 seconds for 200 images.
- **OBJ format** is widely supported, making it easy to view and use the 3D model in different software.
- **The pipeline is reusable** — we can now easily process different sets of images by simply changing the input folder.

The complete pipeline successfully transforms 2D silhouette images into a 3D model, demonstrating that our approach works end-to-end.

5 Conclusion

This week, we successfully created a complete pipeline that covers the entire journey from image loading to 3D OBJ formation.

The key achievements include:

- Integrated all stages (image loading, silhouette extraction, voxel carving, mesh extraction, OBJ export) into a single pipeline.

- Successfully generated a 3D OBJ file from diamond silhouette images.
- The pipeline processes 200 images in under 20 seconds.
- The final OBJ file can be opened and viewed in standard 3D software.

The complete pipeline demonstrates that we can successfully reconstruct a 3D diamond model from 2D silhouette images, completing the main objective of our project.

The pipeline is functional, efficient, and produces accurate results that can be used for visualization and further analysis.